

Melek Moalla

AI Engineering Student | ESPRIT | Tunis, Tunisia

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Portfolio: melek-moalla.vercel.app

EDUCATION & RECOGNITION

ESPRIT - Computer Science Engineering, Artificial Intelligence track | Expected 2028

Ranked first in annual class ranking (2024-25); second (2023-24). NVIDIA certificates in Machine Learning, Deep Learning and Generative AI. LogiXpress: Bal des Projets winner.

EXPERIENCE

Software & AI Intern - MS Solutions Group | Summer 2026

Real-time transaction monitoring with FastAPI, WebSockets and Redis; conversational access to dashboard data through an LLM chatbot.

Software Intern - Telnet | 2024-2025

Built and deployed a Docker-based live metrics dashboard from supervisor specifications.

Web Development Intern - Shamash IT | 2023-2024

Developed user-facing interfaces with HTML, CSS and JavaScript.

SELECTED PROJECTS

Reward Goblin | Python, PyTorch, Stable-Baselines3, Gymnasium

PPO experiments exposing reward specification gaming, with recorded policy/reference comparisons and evaluation on unseen layouts.

Vector | MediaPipe, OpenCV, Python, Win32 API

Webcam gesture interface for moving Windows windows. Deterministic engine and synthetic-motion tests; live-hand tuning remains to be done.

Hydra | TypeScript, React, Electron, MCP

Desktop agent orchestration with seven manager MCP tools, real terminal sessions, headless runs and remote control. No public release yet.

GhostHands | PyTorch, imitation learning

Goal-conditioned pick-and-place policies evaluated on held-out and moved-object layouts. Included checkpoints use synthetic teacher demonstrations.

Plant DNA & crop intelligence platform | TensorFlow, PyTorch, FastAPI, React, Flutter

Multi-module integration of crop classifiers and a retrieval workflow for DNA search; selected for the Bal des Projets deep-learning showcase.

TOOLS & LANGUAGES

Python, TypeScript, JavaScript, C/C++, PHP, SQL | PyTorch, scikit-learn, OpenCV, MediaPipe | FastAPI, React, Docker, Redis, Git, Linux

Languages: French, Arabic, English, Spanish